

NEXUS SAMPLE QA

Nexus Sample Questions and Expected Answers

Use this document after ingesting 'data/raw/nexus_research_bulletin_2025.pdf' into either app.

Best Demo Questions

1. Who is Dr. Elara Vonn and what is ORIN-7?

Expected answer:

Dr. Elara Vonn is the lead researcher behind the Vonn Entropic Compression theory. 'ORIN-7' is her '400-qubit photonic processor' used in 'Halvard, Norway'.

2. Which scientist discovered the Mossad-Chen Fold, and what is its thermal stability?

Expected answer:

'Professor Tariq Mossad' discovered the 'Mossad-Chen Fold', and it remains stable up to '210 degrees C'.

3. What did FRAC-9 achieve in the benchmark results?

Expected answer:

'FRAC-9' ran a '70-billion-parameter language model' on a device with only '512 MB of RAM' at '18 tokens per second', while preserving '96.3%' of the original accuracy on 'NNBS-v4'.

4. What is HYDROS-1 and how efficient is it?

Expected answer:

'HYDROS-1' is a 'room-temperature hydrogen fuel cell' invented by 'Dr. Okafor Nweze', and it achieves '91% efficiency' using a 'bismuth-selenide membrane'.

5. Which city is called the Silicon Fjord, and what initiative reduced energy waste there?

Expected answer:

'Halvard, Norway' is called the 'Silicon Fjord', and the 'Halvard Smart Grid Initiative' reduced energy waste there by '38%'.

Good Entity-Rich Questions

6. Which city uses the Desalination Loop, and what percentage of wastewater does it recycle?

Expected answer:

'Aqaris, Jordan' uses the 'Desalination Loop', and it recycles '99.2%' of wastewater.

7. What is veltranite and why is it important?

Expected answer:

'Veltranite' is a rare earth mineral from 'Veltran, Canada'. It becomes superconductive at '-18 degrees C' when doped with '3% osmium', which makes it important for possible lossless power transmission.

8. What discovery won the Nexus Grand Prize 2025?

Expected answer:

The discovery was 'VELT-OS3', a form of 'veltranite doped with 3% osmium' that achieved superconductivity at '-18 degrees C'. It won the 'Nexus Grand Prize 2025', worth '\$5 million'.

9. What is MCF-S3 and where was it piloted?

Expected answer:

'MCF-S3' is a synthetic variant of the 'Mossad-Chen Fold' protein that can remove 'PFAS chemicals' from water at '99.7% efficiency'. It was piloted at the 'Aqaris Eastern Reservoir'.

10. Who pioneered Neuromorphic Fractal Encoding, and where was it deployed?

Expected answer:

'Dr. Senna Brightwell' pioneered 'Neuromorphic Fractal Encoding (NFE)', and her startup 'BrightCore AI' deployed it across '47 countries' by '2024'.

Strong Multi-Hop Questions

11. Which city hosts the lab of the Bramwell Prize winner, and what machine does that lab use?

Expected answer:

The city is 'Halvard, Norway', and the lab uses 'ORIN-7', a '400-qubit photonic processor'.

12. Which finding could enable lossless transcontinental power transmission, and what exact material change made it possible?

Expected answer:

'FINDING NX-005' could enable lossless transcontinental power transmission. It used 'veltranite doped with 3% osmium', codenamed 'VELT-OS3'.

13. Which scientist worked on deep-sea proteins, and what industrial use came from that work?

Expected answer:

'Professor Tariq Mossad' worked on deep-sea proteins, and that work led to 'MCF-S3', which can neutralize 'PFAS chemicals in water'.

14. Compare Elara Vonn's and Okafor Nweze's work in one answer.

Expected answer:

'Elara Vonn' worked on reversible entropy and quantum coherence using 'ORIN-7', while 'Okafor Nweze' developed 'HYDROS-1', a high-efficiency room-temperature hydrogen fuel cell.

15. Which city hosts the Deep Ocean Data Centre, and which scientist there studied deep-sea protein structures?

Expected answer:

The city is 'Aqaris, Jordan', and the scientist is 'Professor Tariq Mossad'.

Nice Eval-Demo Questions

These are good questions for the eval app because they tend to generate clean retrieval evidence and easy-to-explain judge scores.

- 'What is GraphRAG and how does it improve standard RAG?'
- 'What did FRAC-9 achieve in the benchmark results?'
- 'What discovery won the Nexus Grand Prize 2025?'

Demo Tips

1. Ingest the Nexus PDF first.
2. Ask one direct factual question.
3. Ask one multi-hop question.
4. Show the answer and retrieved chunks.
5. Open the eval dashboard and CloudWatch logs to show judge scores.